Inside the Institution

Visualizing Institutional Collaboration Networks for Meta-Scientific Exploration

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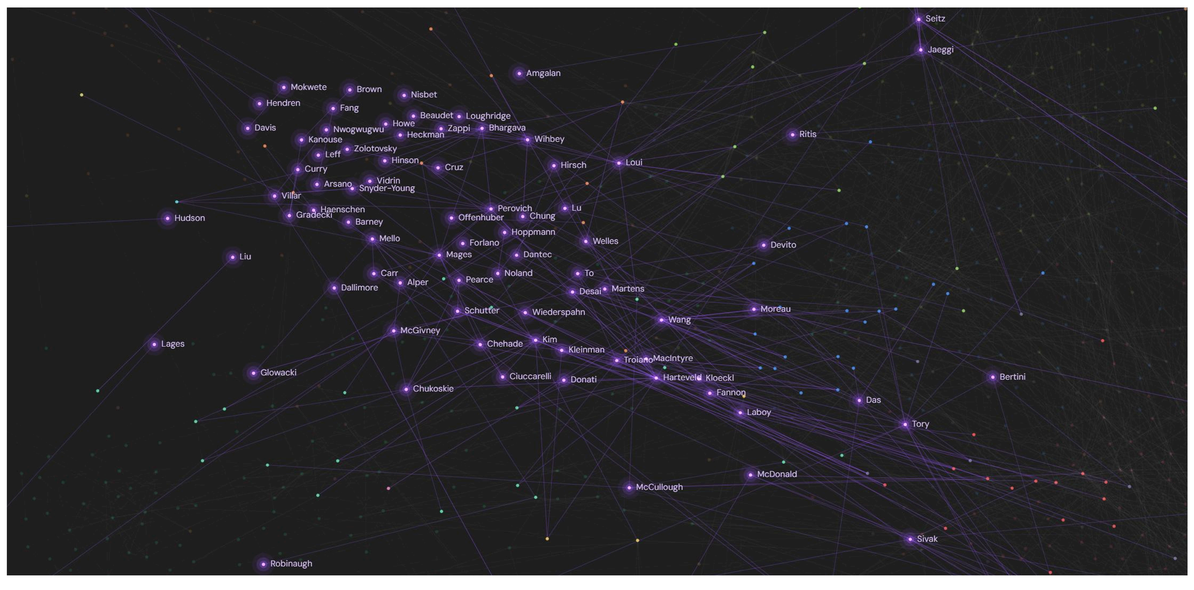
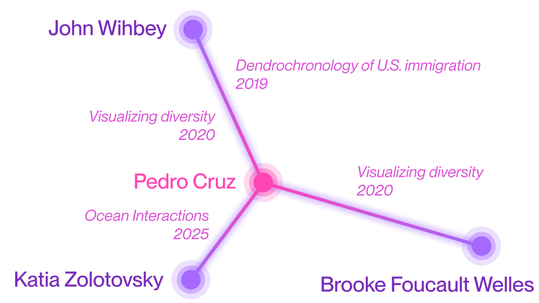


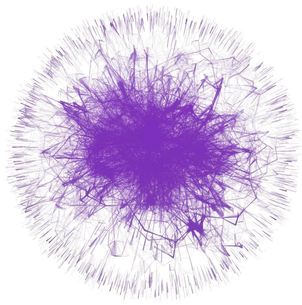
One college's co-authorship network at a research university (tool dataset, D3.js): 80 faculty, 194 cross-college links.

❶ The gap between analysis and exploration

Research universities depend on cross-disciplinary collaboration. Meta-scientists study these patterns through co-authorship networks. Tools like Gephi can render these networks, but the output is static. Researchers can analyze structure, but they cannot easily explore it. This project turns the network into an interactive environment where exploration replaces a fixed view.



Co-authorship as a tie. Two researchers are linked with a shared paper.



A co-authorship network rendered in Gephi, a common network visualization tool.

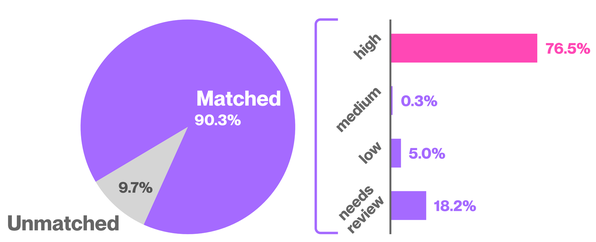
❷ Research questions

RQ1: How might we reveal the hidden **patterns** of intra-institutional co-authorship, and the organizational **structure** that produces them?

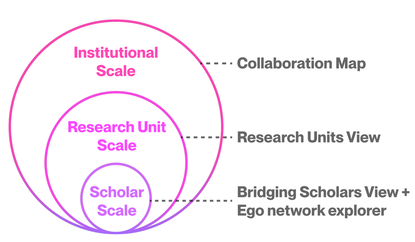
RQ2: How might an **interactive visualization** show collaboration patterns that existing tools cannot show, and that researchers would not think to look for?

❸ From data to design

The project assembled a dataset of 2,737 Northeastern faculty across 11 colleges, 46 departments, and 64 research centers, linked through 3,650 co-authored works. To keep a network of this scale legible, the tool organizes exploration into three views: institution, unit, and scholar.



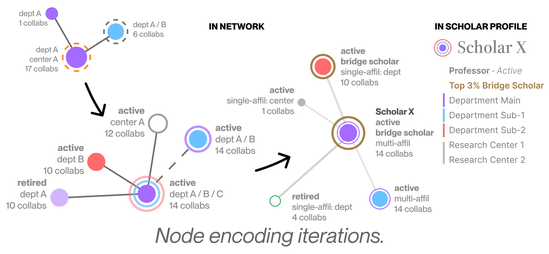
OpenAlex matched 90.3% of Northeastern faculty. Of those, 76.5% returned high-quality publication records; 18.2% were flagged for manual review.



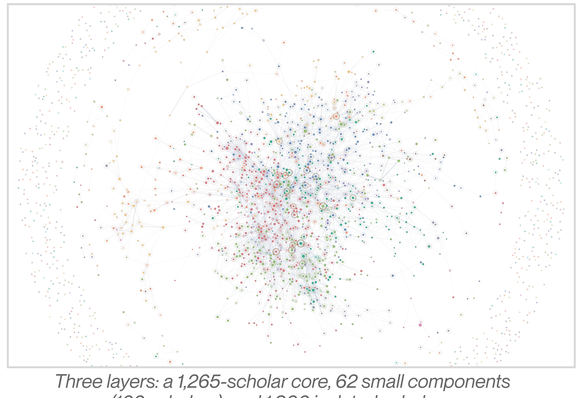
Three views, three scales. Each view answers a different kind of question meta-scientists ask about an institutional collaboration network.

❹ Collaboration map

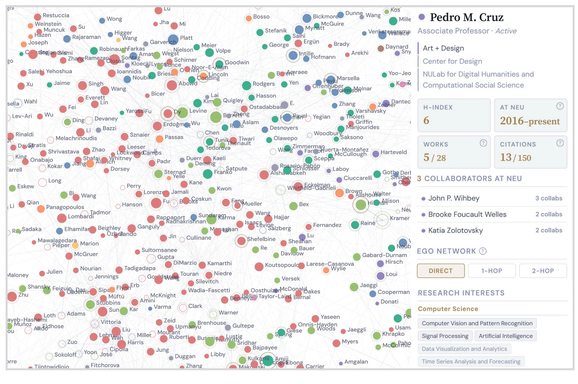
Collaboration Map shows the institution as a single network. Faculty appear as nodes connected by co-authorship. The layout reveals the overall structure: tight clusters of connected scholars, and regions that remain loosely connected or detached.



Node encoding iterations.



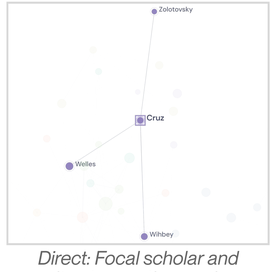
Three layers: a 1,265-scholar core, 62 small components (166 scholars), and 1,306 isolated scholars.



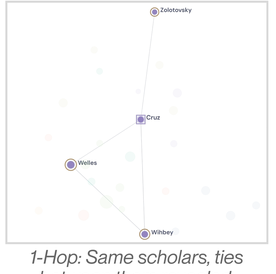
Zoomed in, scholars become searchable.

❺ Ego network explorer

Collaboration Map shows the institution. *Ego Network Explorer* shows what it looks like from one scholar's position inside it.



Direct: Focal scholar and direct co-authors only.



1-Hop: Same scholars, ties between them revealed.

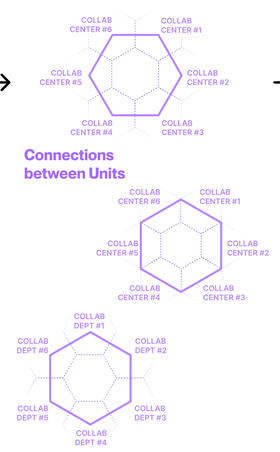
2-Hop: Extended to co-authors of co-authors.

Click any node to recenter on new collaborators within hop(s).

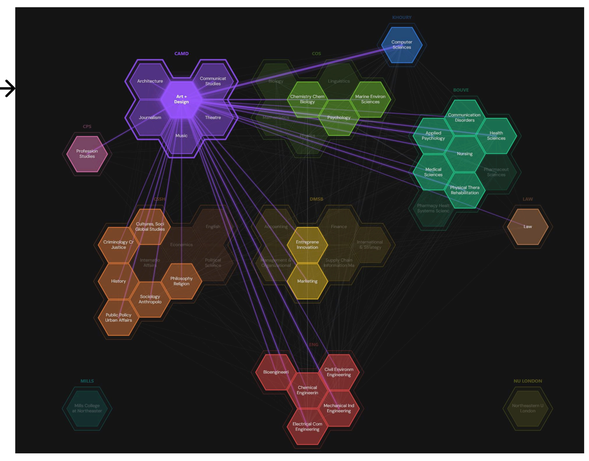
❻ Research units view

Research Units shifts the focus from people to structures. Each department and research center becomes a hexagonal unit, making collaboration within and between them visible.

Inspiration: board-game tiles as modular containers.



Draft: departments flat-topped, research centers pointy-topped.



In the tool: hexagons nested by college, connected by co-authorship.

❼ Bridging scholars view

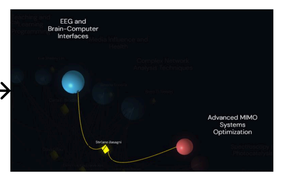
Bridging Scholars focuses on individuals who connect different communities. Each bridging scholar sits in a 3D layout where position reflects the research areas they span.

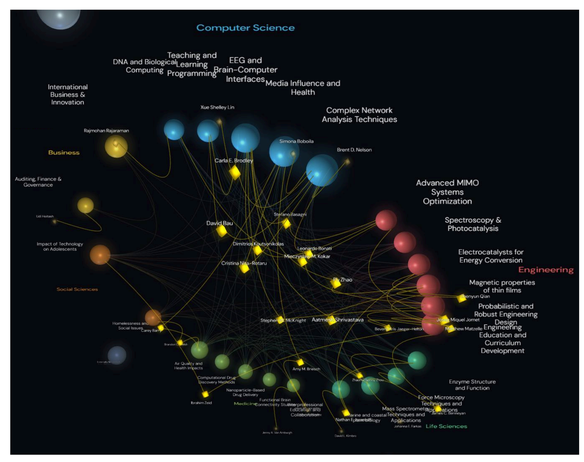
Bridge Score =

$$\frac{\text{cross-department partners}}{\text{total partners}} \times \log_{10}(\text{total partners})$$

A scholar who collaborates widely across departments scores higher than one who collaborates heavily within their own. The log keeps productive but narrow collaborators from dominating.

Inspiration: universities as planetary systems





From planetary systems to a 3D layout: bridging scholars at the center, research topics as spheres.

❽ Visual languages

Each view uses its own grammar: dots for individuals, hexagons for units, planets for bridging scholars. Forcing one grammar across all three would have flattened the question each view is built to answer.

❾ Answers to the RQs

RQ1: The network is **fragmented**. Only **52%** of faculty have internal co-authorship ties. Colleges vary widely in how well-connected they are. A small group of interdisciplinary scholars holds structurally critical positions.

RQ2: Interactive visualization can reveal what analysis alone cannot. Shown by the results of the user study.

❿ User study

Twenty meta-scientists tested the tool. Each session combined free exploration with think-aloud observation, followed by a short questionnaire.

⓫ What they found

Meta-scientists independently surfaced structural patterns across the institution: gaps between colleges, bridge scholars who connect otherwise disconnected communities, disciplinary asymmetries in how research gets published.

> *It's not obvious how much or how little cross-departmental collaboration exists.* **Something like this in the hands of an administrator is super useful.**
>
> *— Participant 07, from the user study*

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